

| parameter                                                                                             | UCI                                                 | BU       | notes                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N.B. when the BU column is left empty, the value used is the same as the one listed in the UCI column |                                                     |          |                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>p.screen</b>                                                                                       |                                                     |          | <b>screen parameters</b>                                                                                                                                                                                                                                                                                                                                                        |
| distFromScreen_inCm:                                                                                  | 75                                                  | 75       |                                                                                                                                                                                                                                                                                                                                                                                 |
| screenWidth_inPixels:                                                                                 | 1280                                                | 1920     |                                                                                                                                                                                                                                                                                                                                                                                 |
| screenHeight_inPixels:                                                                                | 1024                                                | 1080     |                                                                                                                                                                                                                                                                                                                                                                                 |
| screenWidth_inCm:                                                                                     | 40                                                  | 50       |                                                                                                                                                                                                                                                                                                                                                                                 |
| screenHeight_inCm:                                                                                    | 30                                                  | 30       |                                                                                                                                                                                                                                                                                                                                                                                 |
| refreshRate_inHz:                                                                                     | 60                                                  | 120      |                                                                                                                                                                                                                                                                                                                                                                                 |
| contrastResolution                                                                                    | 0.0078                                              |          | minimum possible contrast                                                                                                                                                                                                                                                                                                                                                       |
| <b>p.stim</b>                                                                                         |                                                     |          | <b>general stimulus parameters</b>                                                                                                                                                                                                                                                                                                                                              |
| apertureWidth_inDeg:                                                                                  | 4                                                   |          | "aperture" refers to the gray circle in which the fixation cross and cues are displayed                                                                                                                                                                                                                                                                                         |
| fixationWidth_inDeg:                                                                                  | 0.35                                                |          | fixation cross width                                                                                                                                                                                                                                                                                                                                                            |
| cueWidth_inDeg:                                                                                       | 0.1                                                 |          | cue with in degrees                                                                                                                                                                                                                                                                                                                                                             |
| cueLength_inDeg:                                                                                      | 1                                                   |          | cue length in degrees                                                                                                                                                                                                                                                                                                                                                           |
| cueBuffer_inDeg:                                                                                      | fix                                                 |          | distance b/t fixation center and cue edge                                                                                                                                                                                                                                                                                                                                       |
| BGcolor:                                                                                              | 127                                                 |          | background color when no line texture stim is shown                                                                                                                                                                                                                                                                                                                             |
| cueColorInactive:                                                                                     | 0                                                   |          | color of inactive cues. Cues are inactive when drawn on the screen but not actively directing attention or response                                                                                                                                                                                                                                                             |
| cueColorActive:                                                                                       | 255                                                 |          | color of active cues. Cues are active when directing attention (precue) or response (postcue)                                                                                                                                                                                                                                                                                   |
| fixationColor                                                                                         | [0 191.25 255]                                      |          | color of fixation (default), color of fixation following response 1, color of fixation after response 2                                                                                                                                                                                                                                                                         |
| <b>p.stim.periph</b>                                                                                  |                                                     |          | <b>peripheral stimulus parameters</b>                                                                                                                                                                                                                                                                                                                                           |
| stimDistFromCenter_inDeg:                                                                             | 5                                                   |          | distance from center of screen to center of peripheral stimulus                                                                                                                                                                                                                                                                                                                 |
| ovalMajorAxis_inDeg                                                                                   | 5                                                   |          | length of the peripheral stimulus along its major axis (in degrees)                                                                                                                                                                                                                                                                                                             |
| ovalMinorAxis_inDeg                                                                                   | 5                                                   |          | length of the peripheral stimulus along its minor axis (in degrees)                                                                                                                                                                                                                                                                                                             |
| circleWidth_inDeg                                                                                     | 5                                                   |          | width of a circle whose area equals the area of the peripheral stimulus with major and minor axes as specified above (in degrees)                                                                                                                                                                                                                                               |
| <b>p.stim.gab</b>                                                                                     |                                                     |          | <b>peripheral grating and noise parameters</b>                                                                                                                                                                                                                                                                                                                                  |
| gab.aperture                                                                                          | cosine'                                             |          | grating aperture type                                                                                                                                                                                                                                                                                                                                                           |
| gab.apertureEdgeWidth_inDeg                                                                           | 0.5                                                 |          | grating aperture edge width (in degrees)                                                                                                                                                                                                                                                                                                                                        |
| gab.apertureEdgeWidth_inPix                                                                           | 25                                                  |          | grating aperture edge width (in pixels)                                                                                                                                                                                                                                                                                                                                         |
| gab.rad                                                                                               | 100                                                 |          | grating radius (in pixels), equal to periph.circleWidth_inPix/2 - gab.apertureEdgeWidth_inPix                                                                                                                                                                                                                                                                                   |
| gab.noiseType                                                                                         | filteredSF'                                         |          | type of noise, filtered around grating spatial frequency                                                                                                                                                                                                                                                                                                                        |
| gab.noiseContrast                                                                                     | 0.5                                                 |          | noise contrast                                                                                                                                                                                                                                                                                                                                                                  |
| gab.gratingSF                                                                                         | 1                                                   |          | grating spatial frequency (in cycles per degree)                                                                                                                                                                                                                                                                                                                                |
| gab.orientation                                                                                       | [-45, 45]                                           |          | list of possible grating orientations (in degrees) before tilt thresholding                                                                                                                                                                                                                                                                                                     |
| gab.phase                                                                                             | 0                                                   |          | grating phase                                                                                                                                                                                                                                                                                                                                                                   |
| gab.gratingContrast_min                                                                               | 0.05                                                |          | minimum possible grating contrast (for determining priors only)                                                                                                                                                                                                                                                                                                                 |
| gab.gratingContrast_max                                                                               | 0.2                                                 |          | maximum possible grating contrast (for determining priors only)                                                                                                                                                                                                                                                                                                                 |
| gab.contrastPrior_list                                                                                | [:0.1123 0.1295 0.1493 0.1721 0.2235 0.2903 0.3370] |          | list of 7 possible grating contrasts. This is a pre-defined prior based on data from the cue gab detection experiment                                                                                                                                                                                                                                                           |
| gab.contrast_list                                                                                     | [:0.1123 0.1295 0.1493 0.1721 0.2235 0.2903 0.3370] |          | list of 7 possible grating contrasts                                                                                                                                                                                                                                                                                                                                            |
| <b>p.timing</b>                                                                                       |                                                     |          | <b>duration of each trial stage</b>                                                                                                                                                                                                                                                                                                                                             |
| fixation1Dur_inSecs:                                                                                  | 0.5                                                 |          | initial fixation. If eyetracking, this is the "minimum" duration of this trial stage (but may be longer depending on time needed for fixation acquisition)                                                                                                                                                                                                                      |
| fixation3Dur_inSecs:                                                                                  | 0.15                                                |          | next fixation stage (inactive cues appear)                                                                                                                                                                                                                                                                                                                                      |
| precueDur_inSecs:                                                                                     | 0.05                                                |          | activation of the precue (active cue changes color)                                                                                                                                                                                                                                                                                                                             |
| prePeriphStimDur_inSecs:                                                                              | 0.25                                                |          | next fixation stage (all cues return to inactive status)                                                                                                                                                                                                                                                                                                                        |
| periphStimDur_inSecs:                                                                                 | 0.25                                                |          | presentation of the peripheral noisy grating stimuli                                                                                                                                                                                                                                                                                                                            |
| fixation4Dur_inSecs:                                                                                  | 0.25                                                |          | next fixation stage (all peripheral stimuli removed, only fixation and inactive cues remain)                                                                                                                                                                                                                                                                                    |
| postcueDur_inSecs:                                                                                    | 5                                                   |          | activation of the postcue (active cue changes color). This is the "maximum" duration for this trial stage; in practice, this trial stage ends as soon as the subject enters their response. If no response is entered within the maximum time limit, the trial stage ends automatically                                                                                         |
| postcue2Dur_inSecs:                                                                                   | 5                                                   |          | activation of the 2nd postcue (active cue changes color). This is the "maximum" duration for this trial stage; in practice, this trial stage ends as soon as the subject enters their response. If no response is entered within the maximum time limit, the trial stage ends automatically                                                                                     |
| ITIDur_inSecs:                                                                                        | 0.5                                                 |          | time between keypress on trial N and onset of initial fixation for trial N+1                                                                                                                                                                                                                                                                                                    |
| <b>p.eyetracker.trial</b>                                                                             |                                                     |          | <b>fixation acquisition prior to each trial</b>                                                                                                                                                                                                                                                                                                                                 |
| fixationRadius_inDeg:                                                                                 | 2                                                   |          | radius of the circle in which fixation must remain during fixation acquisition                                                                                                                                                                                                                                                                                                  |
| fixationDur_inSecs:                                                                                   | 0.5                                                 |          | length of time in which fixation must remain in the fixation region during fixation acquisition                                                                                                                                                                                                                                                                                 |
| timeoutDur_inSecs:                                                                                    | 10                                                  |          | if fixation is not acquired within this time limit, the subject is shown a text prompt and fixation acquisition resets. After 2 consecutive timeouts, the subject is prompted to get the experimenter to help with eyetracker calibration                                                                                                                                       |
| <b>p.eyetracker.block</b>                                                                             |                                                     |          | <b>fixation acquisition prior to each block (more stringent than pre-trial fixation)</b>                                                                                                                                                                                                                                                                                        |
| fixationRadius_inDeg:                                                                                 | 1                                                   |          | radius of the circle in which fixation must remain during fixation acquisition                                                                                                                                                                                                                                                                                                  |
| fixationDur_inSecs:                                                                                   | 1                                                   |          | length of time in which fixation must remain in the fixation region during fixation acquisition                                                                                                                                                                                                                                                                                 |
| timeoutDur_inSecs:                                                                                    | 10                                                  |          | if fixation is not acquired within this time limit, the subject is shown a text prompt and fixation acquisition resets. After 2 consecutive timeouts, the subject is prompted to get the experimenter to help with eyetracker calibration                                                                                                                                       |
| <b>p.kb</b>                                                                                           |                                                     |          | <b>keyboard entry of responses</b>                                                                                                                                                                                                                                                                                                                                              |
| exitKey:                                                                                              |                                                     | 'ESCAPE' | key used to exit the experiment                                                                                                                                                                                                                                                                                                                                                 |
| respKeys:                                                                                             | { '1' '2@' '9' '0' }                                | 'ESCAPE' | valid keys for response entry during the experiment. Meanings of the keys are as follows:<br>1 - test stronger than reference, tilt counterclockwise from vertical<br>2 - reference stronger than test, tilt counterclockwise from vertical<br>9 - reference stronger than test, tilt clockwise from vertical<br>0 - test stronger than reference, tilt clockwise from vertical |
| resp2Keys:                                                                                            | { '3#' '8*' }                                       | 'ESCAPE' | valid keys for response 2 entry during the experiment. Meanings of the keys are as follows:<br>3 - guessed the direction of the tilt<br>8 - saw the direction of the tilt                                                                                                                                                                                                       |
| <b>p.auditoryFB</b>                                                                                   |                                                     |          | <b>auditory feedback during practice</b>                                                                                                                                                                                                                                                                                                                                        |
| toneFreq:                                                                                             | [523, 784]                                          |          | tone frequency (in Hz) following incorrect and correct responses, respectively                                                                                                                                                                                                                                                                                                  |
| toneDur_inSecs:                                                                                       | 0.25                                                |          | tone duration in seconds                                                                                                                                                                                                                                                                                                                                                        |
| <b>p.quest</b>                                                                                        |                                                     |          | <b>QUEST thresholding</b>                                                                                                                                                                                                                                                                                                                                                       |
| pThreshold:                                                                                           | 0.75                                                |          | target value of p(correct) for thresholding in the oval orientation discrimination task                                                                                                                                                                                                                                                                                         |
| tGuess:                                                                                               | -0.1249                                             |          | initial guess of the tilt threshold, based on previous pilot data collection                                                                                                                                                                                                                                                                                                    |
| tGuessSD:                                                                                             | 3                                                   |          | SD of tGuess, using default / recommended value                                                                                                                                                                                                                                                                                                                                 |
| beta:                                                                                                 | 3.5                                                 |          | slope of the psychometric function, using default / recommended value                                                                                                                                                                                                                                                                                                           |
| delta:                                                                                                | 0.039                                               |          | fraction of trials on which the subject presses blindly (determines upper asymptote of the psychometric function). Based on previous pilot data collection                                                                                                                                                                                                                      |
| gamma:                                                                                                | 0.5                                                 |          | chance rate of p(correct)                                                                                                                                                                                                                                                                                                                                                       |
| grain:                                                                                                | 0.01                                                |          | quantization (step size) of the internal table, using default / recommended value                                                                                                                                                                                                                                                                                               |
| range:                                                                                                | 5                                                   |          | the intensity difference between the largest and smallest intensity that the internal table can store, using default / recommended value                                                                                                                                                                                                                                        |